



Chapter 9

Using Telephone and Cable Networks for Data Transmission

9-1 TELEPHONE NETWORK

*Telephone networks use circuit switching. The telephone network had its beginnings in the late 1800s. The entire network, which is referred to as the **plain old telephone system (POTS)**, was originally an analog system using analog signals to transmit voice.*

Topics discussed in this section:

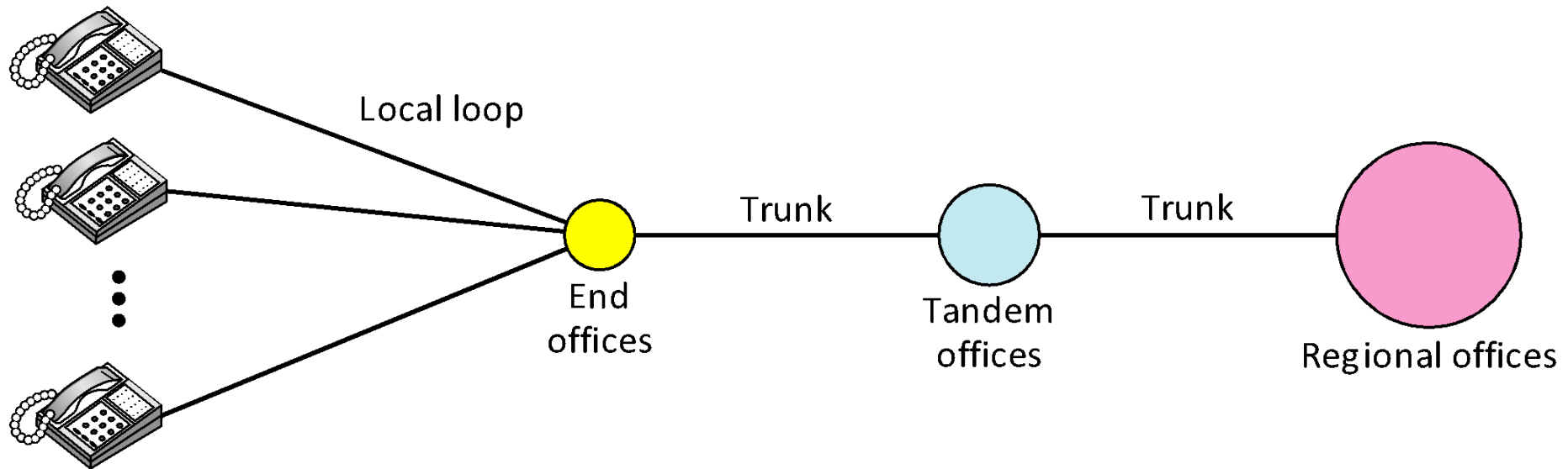
Major Components

LATAs

Signaling

Services Provided by Telephone Networks

Figure 9.1 *A telephone system*



Note

Intra-LATA services are provided by local exchange carriers.
Since 1996, there are two types of LECs: incumbent local exchange carriers and competitive local exchange carriers.

Figure 9.2 *Switching offices in a LATA*

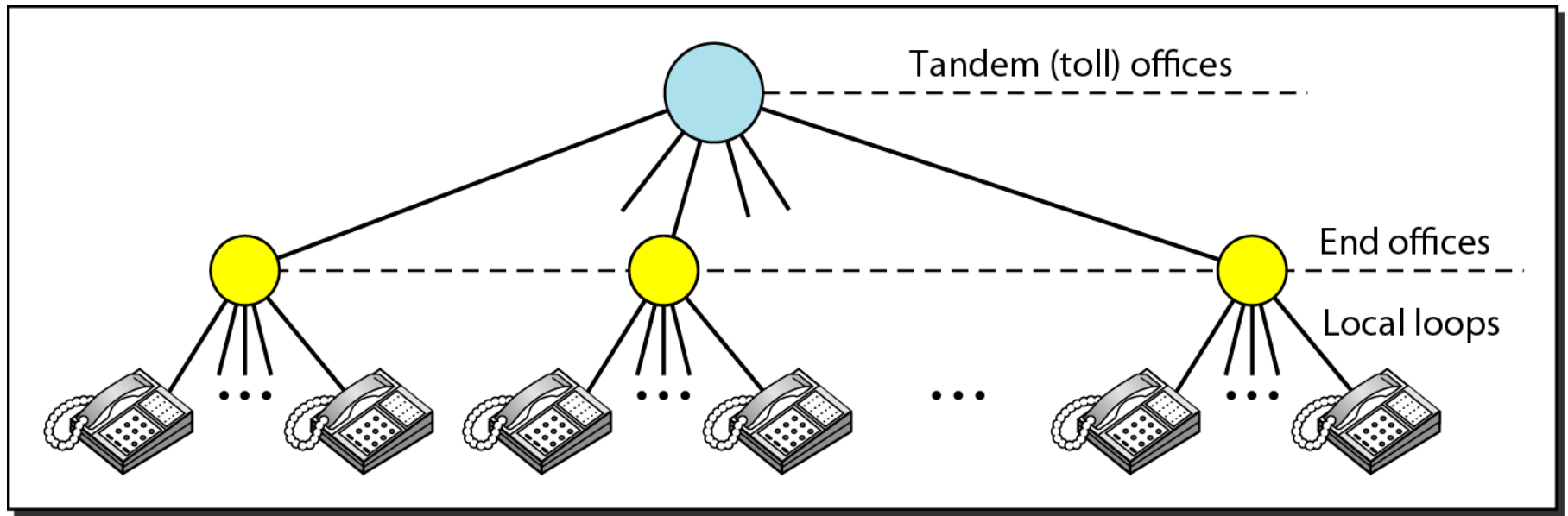
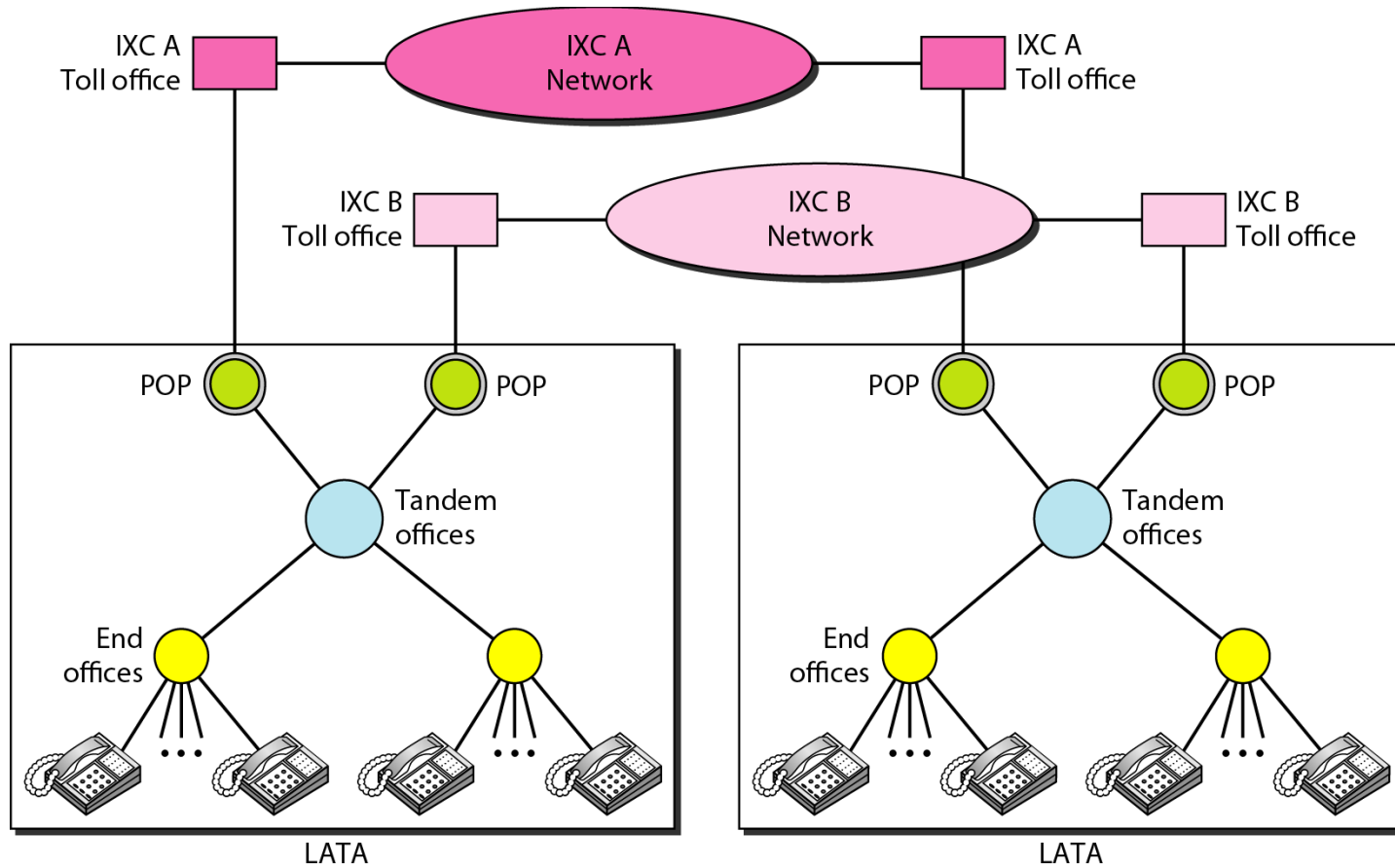


Figure 9.3 *Point of presences (POPs)*



Note

The tasks of data transfer and signaling are separated in modern telephone networks: data transfer is done by one network, signaling by another.

Figure 9.4 *Data transfer and signaling networks*

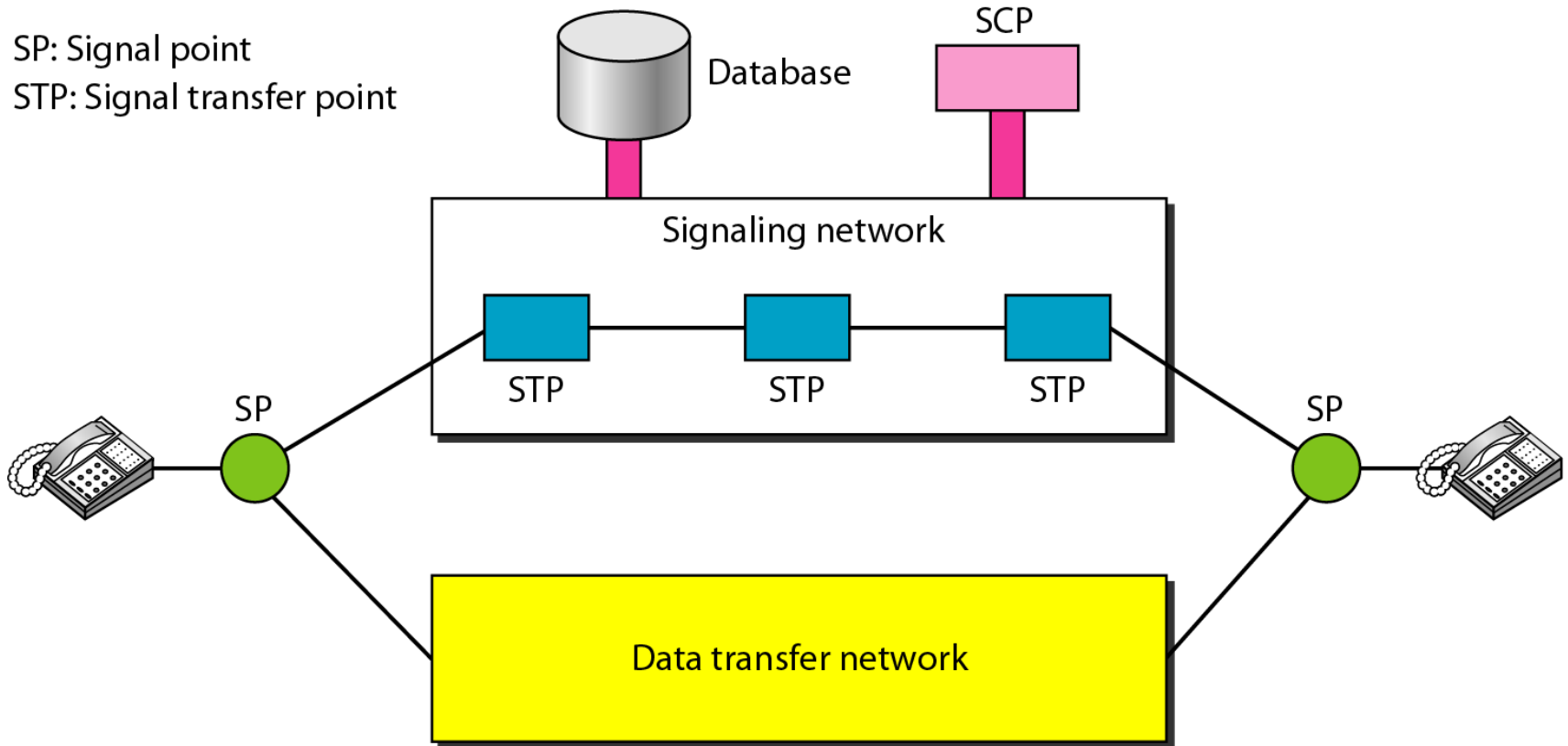
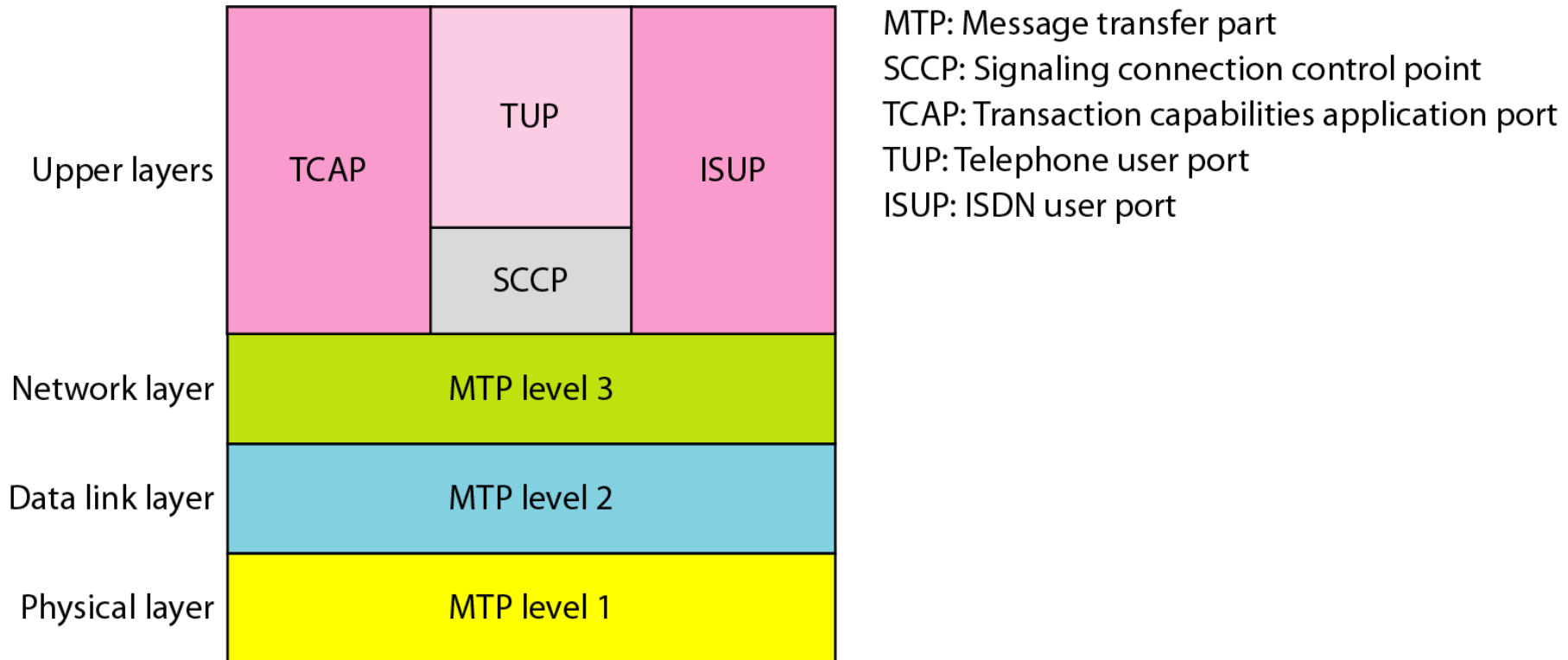


Figure 9.5 *Layers in SS7*



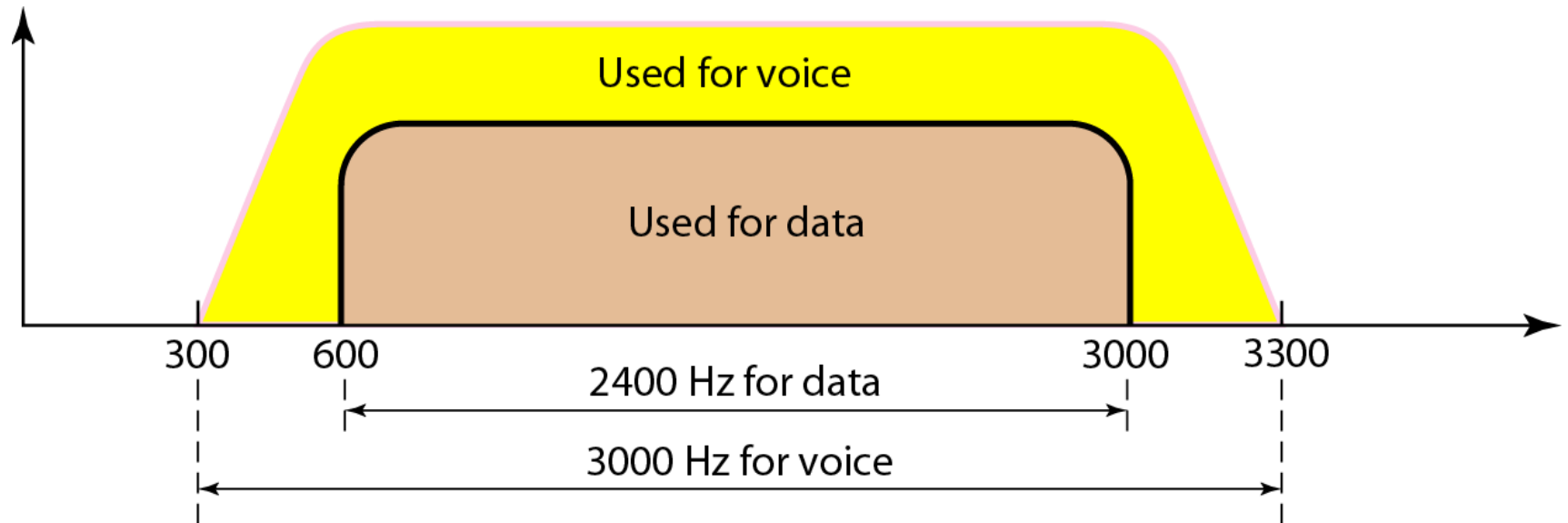
9-2 DIAL-UP MODEMS

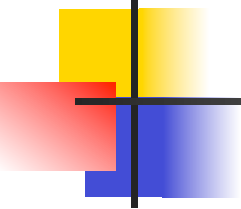
Traditional telephone lines can carry frequencies between 300 and 3300 Hz, giving them a bandwidth of 3000 Hz. All this range is used for transmitting voice, where a great deal of interference and distortion can be accepted without loss of intelligibility.

Topics discussed in this section:

Modem Standards

Figure 9.6 *Telephone line bandwidth*





Note

Modem
stands for modulator/demodulator.

Figure 9.7 *Modulation/demodulation*

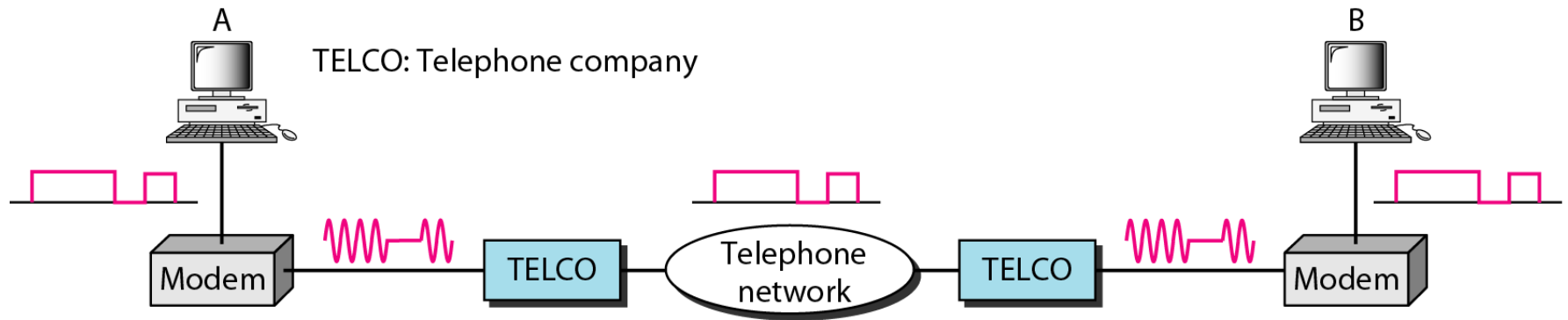
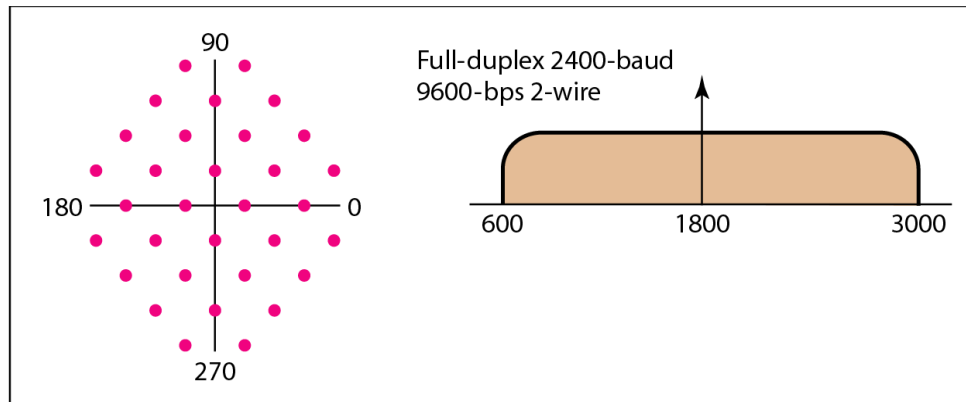
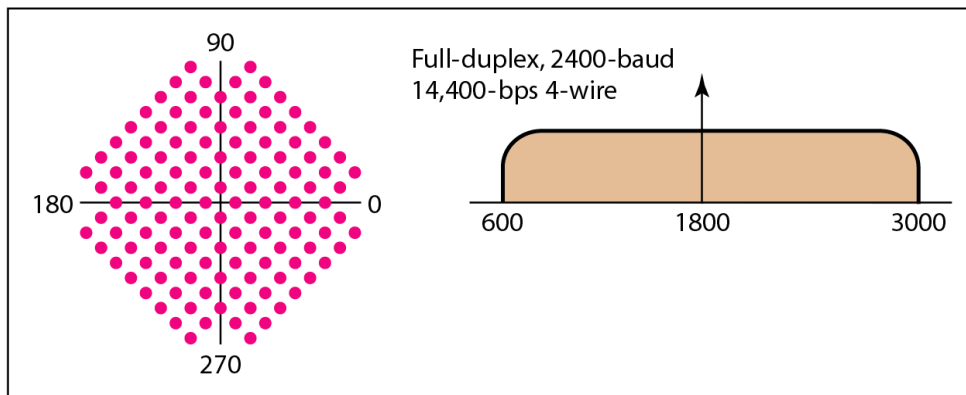


Figure 9.8 *The V.32 and V.32bis constellation and bandwidth*

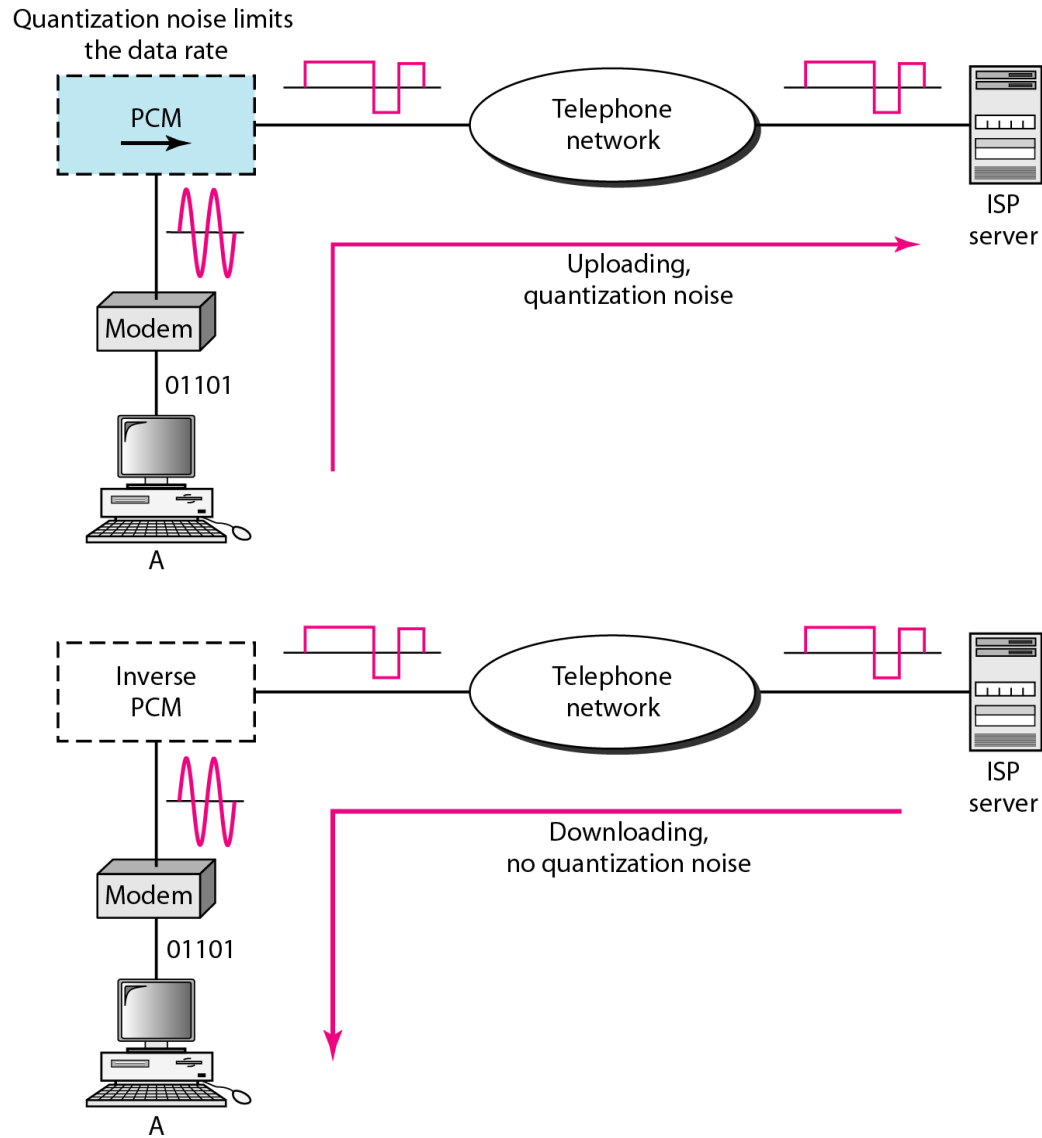


a. Constellation and bandwidth for V.32



b. Constellation and bandwidth for V.32bis

Figure 9.9 *Uploading and downloading in 56K modems*



9-3 DIGITAL SUBSCRIBER LINE

*After traditional modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet. **Digital subscriber line (DSL)** technology is one of the most promising for supporting high-speed digital communication over the existing local loops.*

Topics discussed in this section:

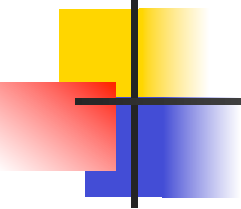
ADSL

ADSL Lite

HDSL

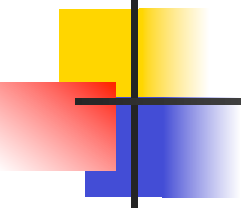
SDSL

VDSL



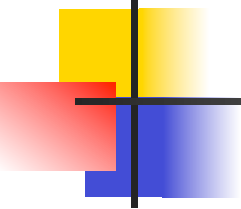
Note

ADSL is an asymmetric communication technology designed for residential users; it is not suitable for businesses.



Note

**The existing local loops can handle
bandwidths up to 1.1 MHz.**



Note

**ADSL is an adaptive technology.
The system uses a data rate
based on the condition of
the local loop line.**

Figure 9.10 *Discrete multitone technique*

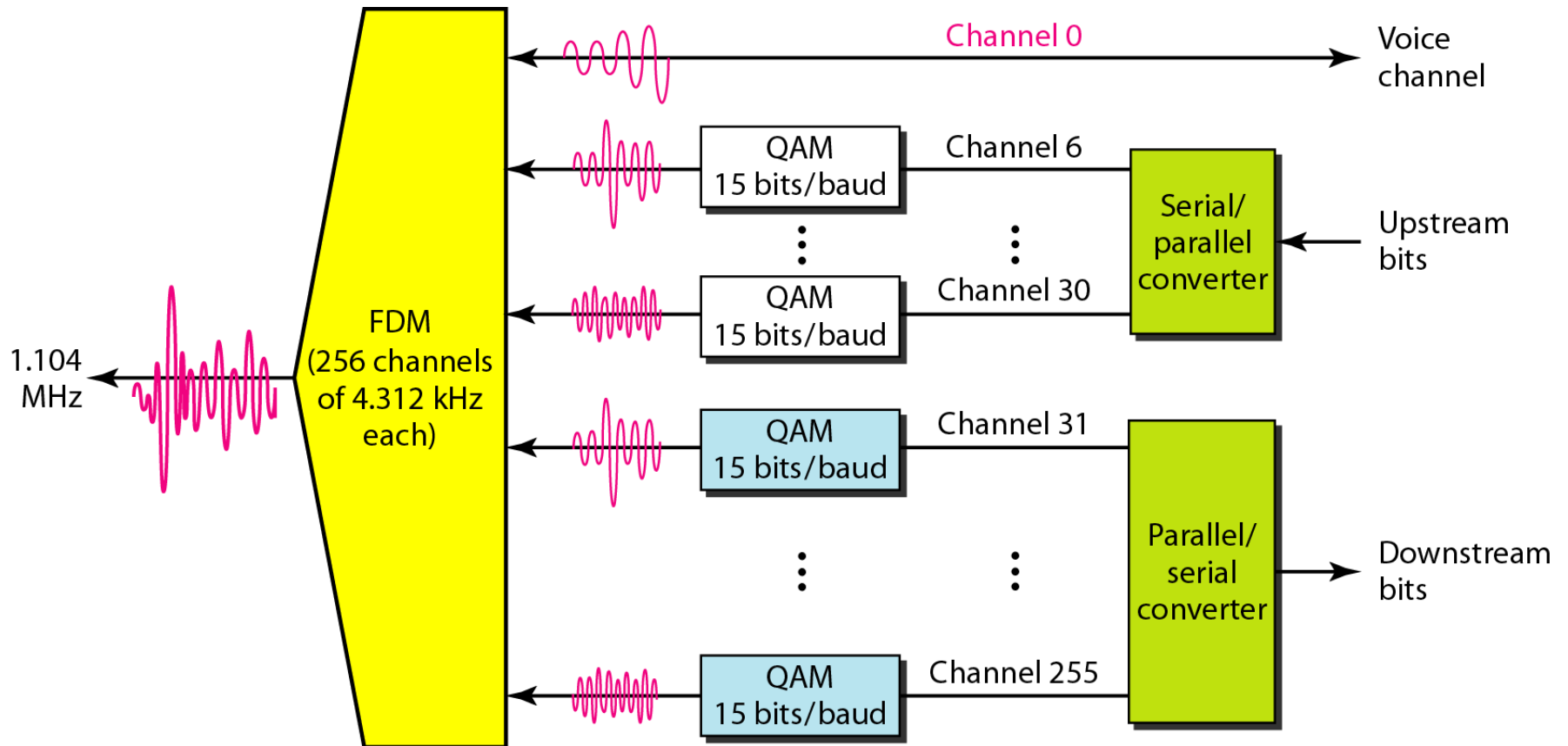


Figure 9.11 *Bandwidth division in ADSL*

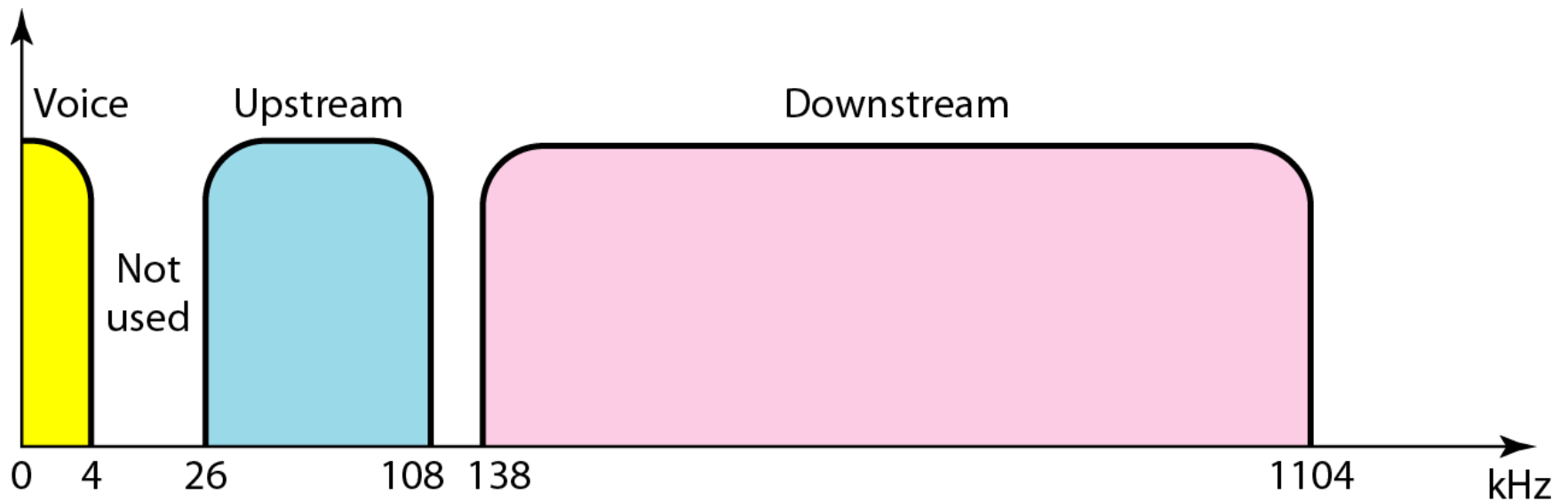


Figure 9.12 *ADSL modem*

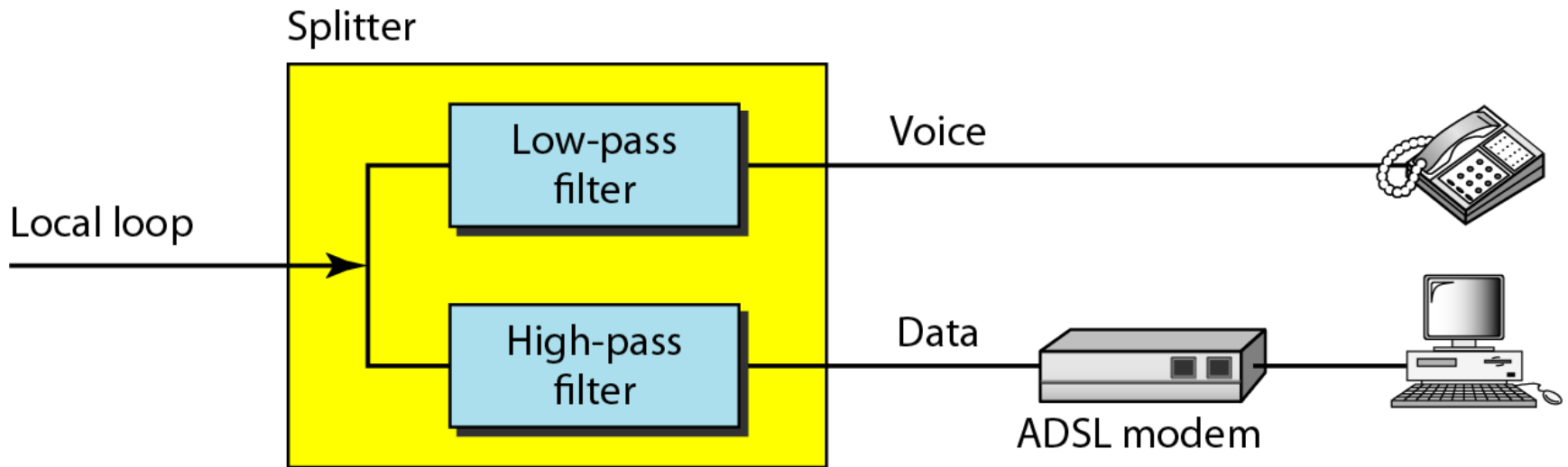


Figure 9.13 *DSLAM*

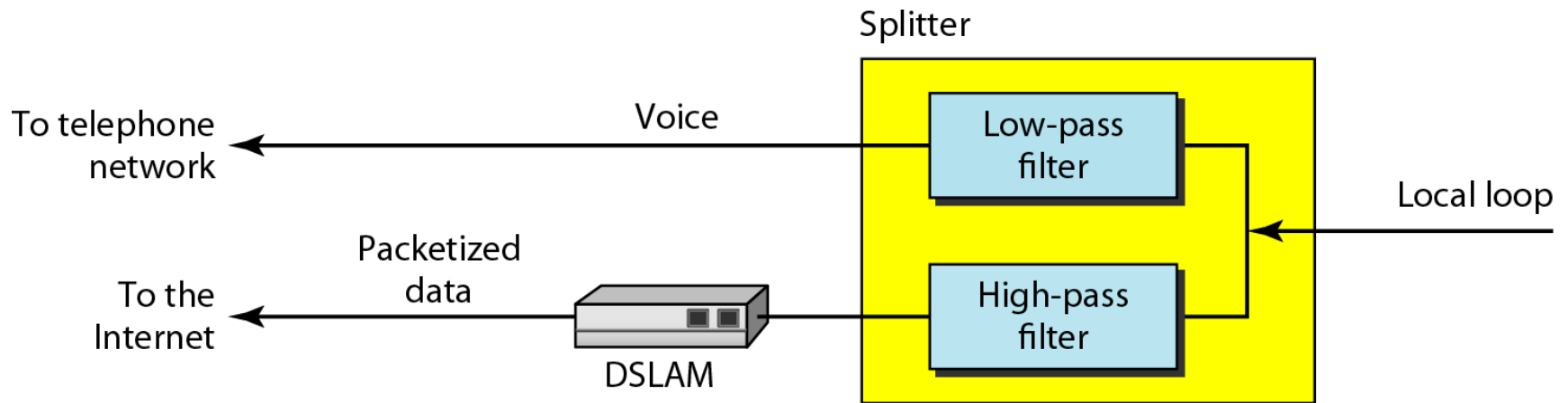


Table 9.2 *Summary of DSL technologies*

<i>Technology</i>	<i>Downstream Rate</i>	<i>Upstream Rate</i>	<i>Distance (ft)</i>	<i>Twisted Pairs</i>	<i>Line Code</i>
ADSL	1.5–6.1 Mbps	16–640 kbps	12,000	1	DMT
ADSL Lite	1.5 Mbps	500 kbps	18,000	1	DMT
HDSL	1.5–2.0 Mbps	1.5–2.0 Mbps	12,000	2	2B1Q
SDSL	768 kbps	768 kbps	12,000	1	2B1Q
VDSL	25–55 Mbps	3.2 Mbps	3000–10,000	1	DMT

9-4 CABLE TV NETWORKS

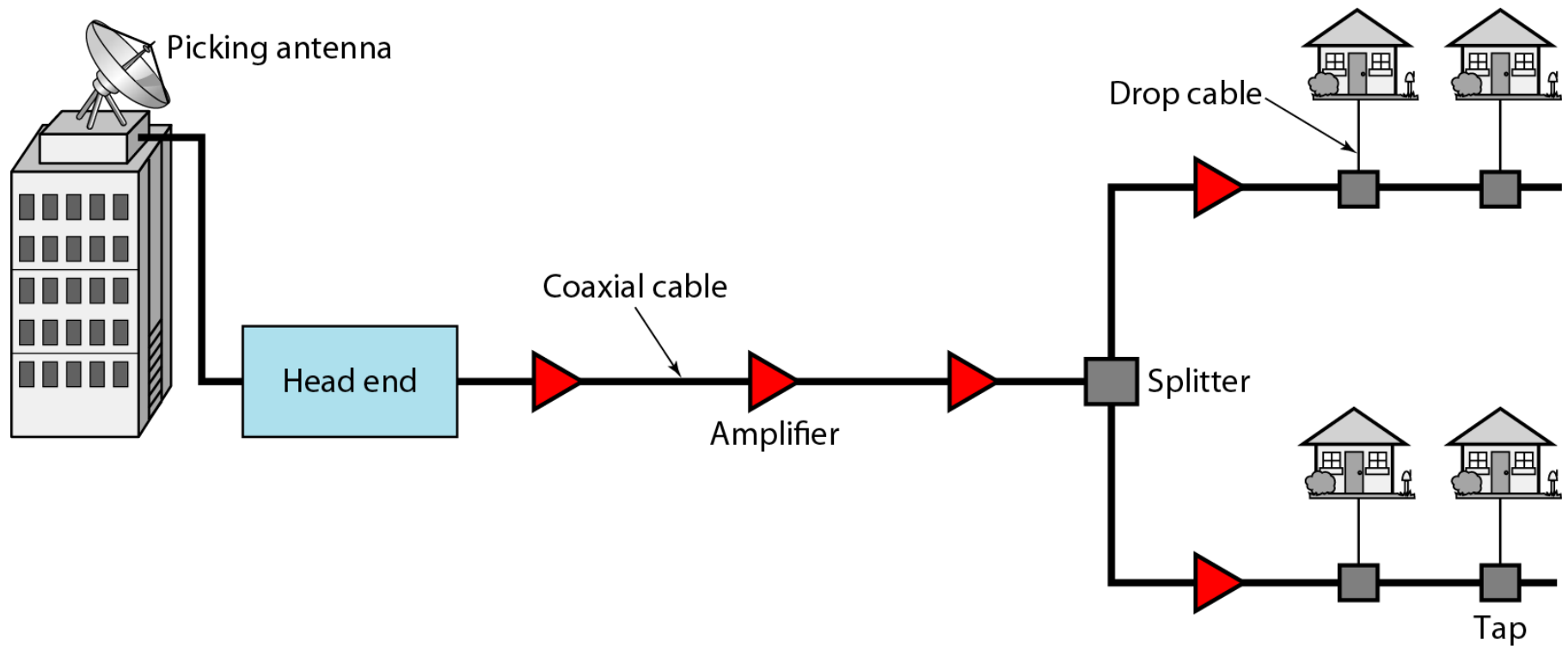
*The **cable TV network** started as a video service provider, but it has moved to the business of Internet access. In this section, we discuss cable TV networks per se; in Section 9.5 we discuss how this network can be used to provide high-speed access to the Internet.*

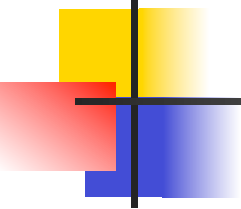
Topics discussed in this section:

Traditional Cable Networks

Hybrid Fiber-Coaxial (HFC) Network

Figure 9.14 *Traditional cable TV network*

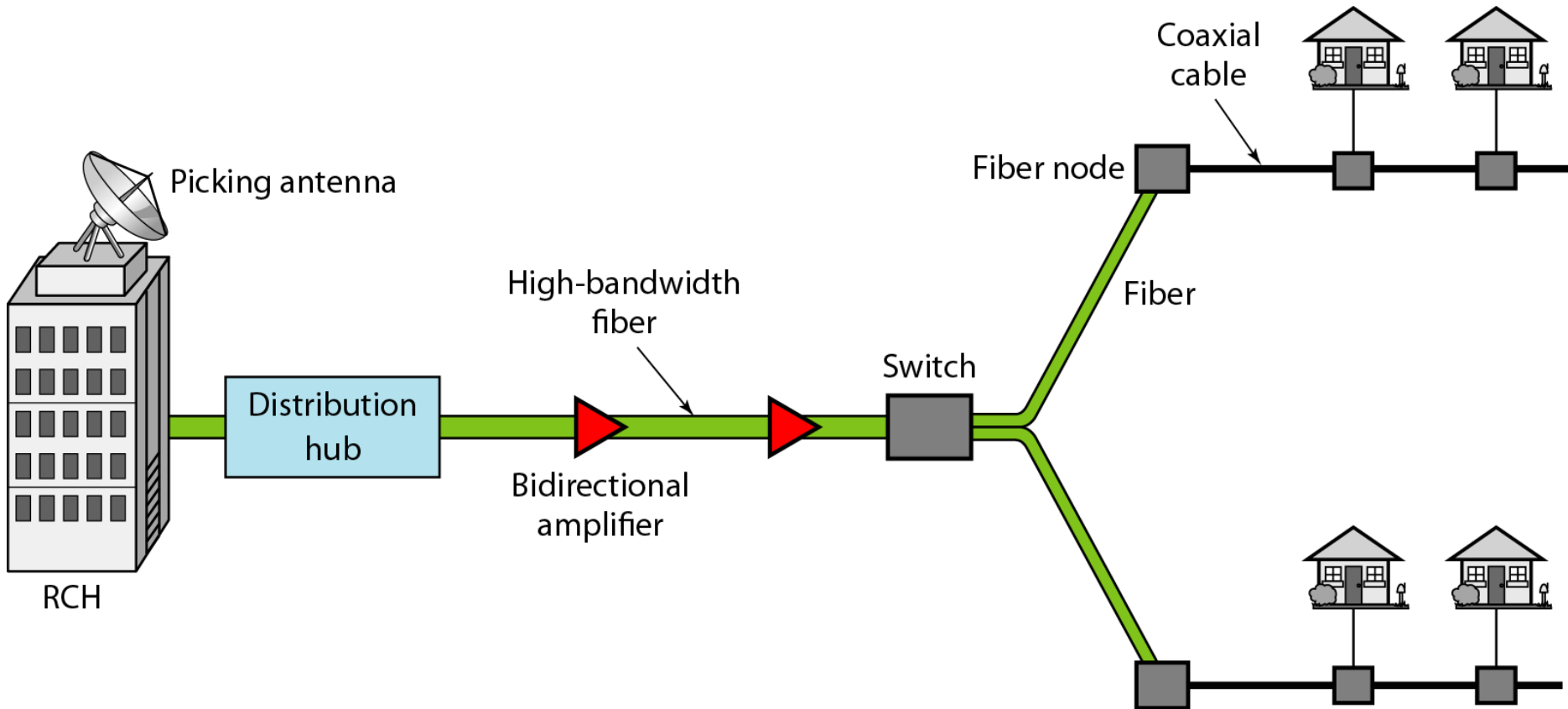


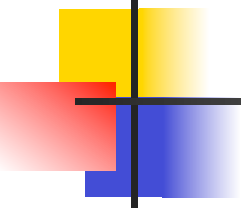


Note

Communication in the traditional cable TV network is unidirectional.

Figure 9.15 *Hybrid fiber-coaxial (HFC) network*





Note

Communication in an HFC cable TV network can be bidirectional.

9-5 CABLE TV FOR DATA TRANSFER

Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer. In this section, we briefly discuss this technology.

Topics discussed in this section:

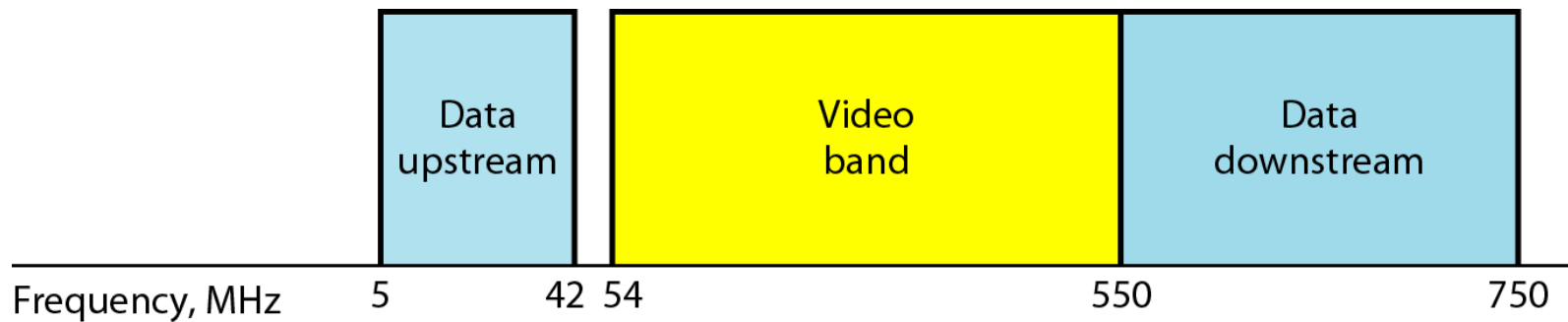
Bandwidth

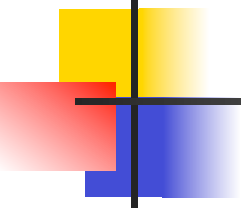
Sharing

CM and CMTS

Data Transmission Schemes: DOCSIS

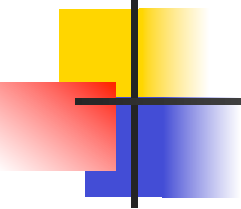
Figure 9.16 *Division of coaxial cable band by CATV*





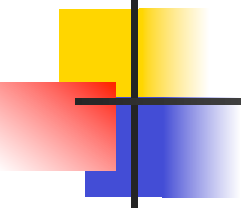
Note

Downstream data are modulated using the 64-QAM modulation technique.



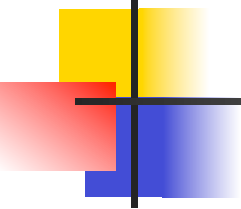
Note

**The theoretical downstream data rate
is 30 Mbps.**



Note

Upstream data are modulated using the QPSK modulation technique.



Note

**The theoretical upstream data rate
is 12 Mbps.**

Figure 9.17 *Cable modem (CM)*

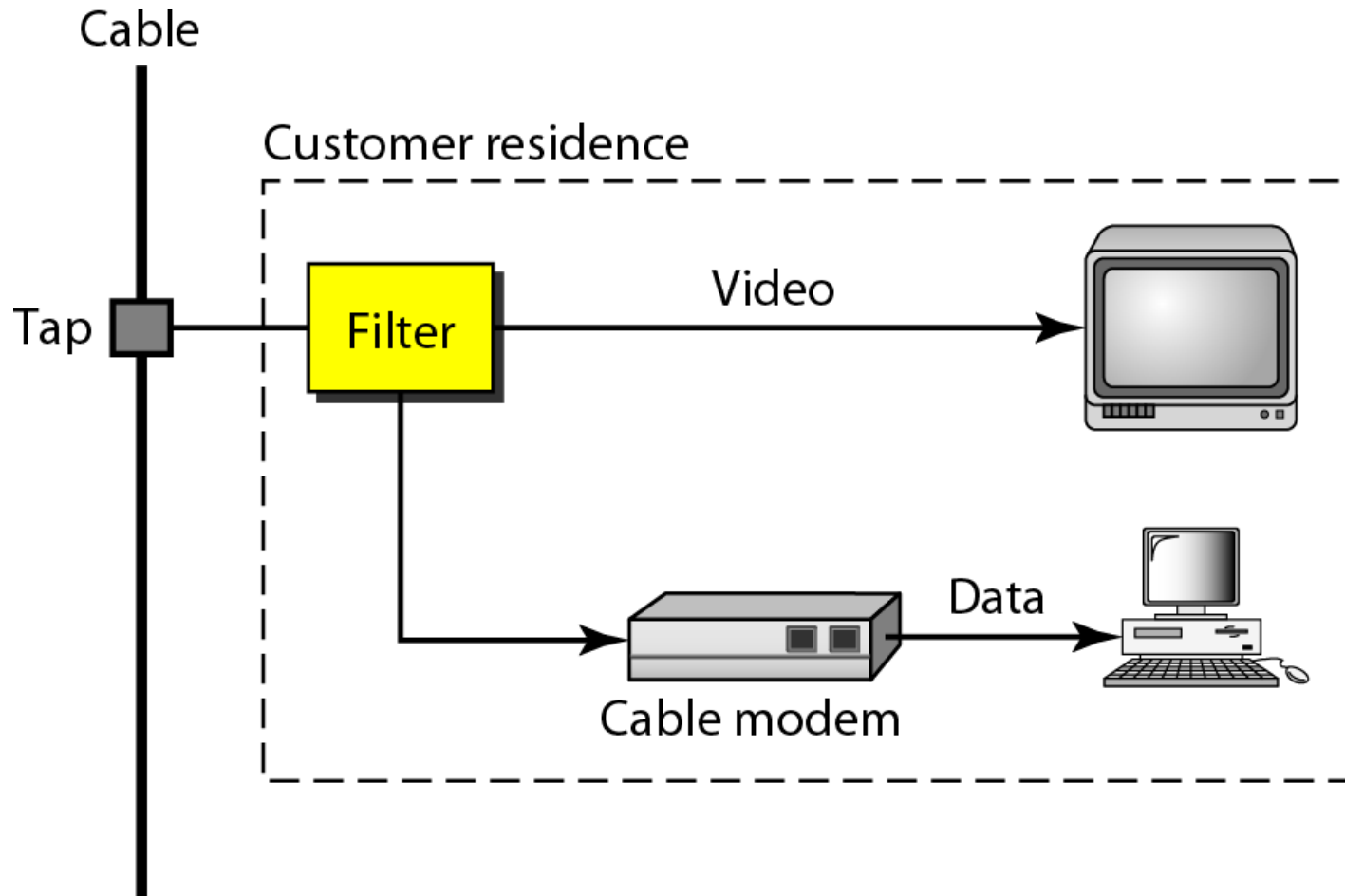


Figure 9.18 *Cable modem transmission system (CMTS)*

